

Autonomous Auction Bidding Agents

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Abstract—This project investigates autonomous bidding agents in a simulated multi-agent auction market using repeated first-price auction rounds. Five strategies were implemented in Python: Random, Conservative, Aggressive, Sniper and Adaptive. The experiments evaluated strategy performance under different levels of competition, information availability and market noise using profit, win rate and bidding efficiency. Results show that no single strategy dominated all metrics. The Conservative Agent achieved the highest average profit, while the Adaptive Agent achieved the highest win rate but lower efficiency. Memory access improved the Adaptive Agent's competitiveness but reduced profit by encouraging overbidding. Overall, the findings show that autonomous bidding agents should be evaluated using multiple metrics rather than win rate alone.

Keywords—autonomous agents, multi-agent systems, auction simulation, bidding strategies, adaptive agents, market noise.

I. INTRODUCTION

Autonomous agents are systems that perceive their environment and select actions to achieve defined objectives [1]. In multi-agent auction environments, agents must also compete with other agents when deciding how much to bid for resources [2], [4].

This project investigates autonomous bidding agents in a simulated auction market. The simulation uses repeated auction rounds where agents bid for generated items using different strategies, including rule-based agents and an adaptive agent that changes its behaviour using previous auction outcomes.

The aim is to compare how bidding strategy, competition, information availability and market noise affect agent performance. Performance is evaluated using profit, win rate, bidding efficiency and failed bids.

II. BACKGROUND

Auction-based systems are used in areas such as electronic commerce, online advertising, cloud resource allocation and spectrum licensing [4], [5]. In these markets, autonomous bidding agents often make decisions with limited information, uncertain item values and competition from other agents [2], [6], [7]. Therefore, the design of a bidding strategy can strongly affect profit, win rate and long-term performance.

This project investigates the following questions:

1. How do different bidding strategies affect agent profitability and performance?
2. How does performance change as market competition increases?
3. Can an adaptive agent improve its behaviour using previous auction outcomes?
4. How does access to market information affect agent decision-making?

5. How does market noise affect bidding performance and stability?

The simulation results are analysed using repeated runs, statistical summaries and visualisations. The main measures are profit, win rate, bidding efficiency, failed bids and performance stability.

III. RELATED LITERATURE

Autonomous agents are systems that perceive their environment and choose actions to achieve goals [1]. In multi-agent systems, each agent's outcome is affected by the actions of others, making the environment competitive and dynamic [2]. This is directly relevant to auction markets, where bidding agents must consider item value and competitor behaviour.

Auction theory provides the basis for this project. Vickrey's work on auctions and sealed tenders helped formalise the study of auction mechanisms [3], while later work examines auction formats, incentives and strategic behaviour [4]. In first-price auctions, the winner pays their own bid, so agents must balance winning against avoiding overpayment. This trade-off connects directly to the project's metrics: profit, win rate and bidding efficiency.

Software agents have also been studied in electronic commerce. Agent-mediated commerce can automate tasks such as comparison, negotiation and purchasing [5]. The Trading Agent Competition further showed how autonomous bidding agents can be evaluated in simulated competitive markets [6], [7]. This supports the use of a simulation environment for analysing bidding behaviour under uncertainty.

Finally, learning is relevant to adaptive agent design. Reinforcement learning is based on agents improving through feedback from previous actions [8]. Although this project does not implement full reinforcement learning, the Adaptive Agent uses auction memory to adjust its aggressiveness, allowing simple learning behaviour to be compared with fixed strategies.

IV. METHODOLOGIES

The simulation tested five autonomous bidding strategies: Random, Conservative, Aggressive, Sniper and Adaptive. These agents represent different levels of risk, competitiveness, timing behaviour and access to market information, which are important factors in auction-based multi-agent systems [2], [4], [6]. All agents were evaluated using the same environment so their profit, win rate and bidding efficiency could be compared fairly.

A. Random Agent

The Random Agent was used as the baseline. It does not use memory or a fixed bidding pattern. Instead, it selects a random valid bid based on the current highest bid, available balance and perceived item value. This provides a simple non-strategic comparison point for the other agents.

B. Conservative Agent

The Conservative Agent follows a low-risk strategy. It only increases the current highest bid by a small amount and avoids bidding close to the perceived item value. This helps preserve profit margins but also makes the agent less likely to win against more aggressive competitors. It therefore represents a risk-averse strategy that prioritises efficiency over frequent wins [4].

C. Aggressive Agent

The Aggressive Agent places larger bid increases above the current highest bid. This makes it more competitive and increases its chance of winning auctions. However, because first-price auction winners pay their own bid, a high win rate does not always lead to high efficiency [3], [4]. The Aggressive Agent is therefore useful for testing the trade-off between winning often and preserving profit.

D. Sniper Agent

The Sniper Agent uses a late-entry strategy. It waits until the later stage of the auction before placing a bid, aiming to avoid unnecessary early competition. When it does bid, it increases the current highest bid by a larger amount. This can work if the price is still low, but it is risky because the agent may lose opportunities if the current bid is already too high.

E. Adaptive Agent

The Adaptive Agent is the main learning-based strategy. It uses auction memory to adjust its bidding behaviour based on previous outcomes, reflecting the idea that agents can improve decisions through feedback [8]. The memory stores winning bids, losing bids, market prices, estimated pressure, volatility and competition level.

The Adaptive Agent starts with a bid based on the perceived item value, then modifies it using memory-based factors. For example, it may bid more aggressively when the market appears competitive or reduce overpayment when recent winning bids are high. It also changes its aggressiveness depending on previous wins and losses. This allows it to respond to changing market conditions, although poor use of memory can still cause overbidding or low efficiency [1], [8].

V. IMPLEMENTATION

The project was implemented in Python using an object-oriented structure. Separate classes were created for the auction environment, auction items, agent behaviours and auction memory. This made the simulation easier to extend, as new agents and experimental conditions could be added without changing the whole system.

A. Auction Environment Design

The auction environment controls the full simulation process. In each auction round, it generates an item, asks agents to submit bids, validates bids, selects the winner, calculates profit and stores the results. Each auction follows a first-price structure where the winner pays the winning bid [3], [4]. However, the implementation updates the current highest bid during the round, so later agents must exceed the current highest valid bid. To reduce ordering bias, agent order is randomly shuffled in every round.

A bid is valid only if it is greater than zero, greater than the current highest bid and within the agent's remaining balance. If no valid bids are submitted, the round ends without a

winner. If multiple agents have the same highest bid, the winner is selected randomly. Profit is calculated as:

$$\text{profit} = \text{item.true_value} - \text{winning_bid} \quad (1)$$

This means agents may make decisions using perceived value, but their final profit is evaluated using the true value.

B. Auction Items and Market Noise

Each auction item stores a true_value and a perceived_value. Under normal conditions, both values are the same. In the market noise experiment, the perceived value is changed using low, medium or high random noise. Agents bid using the perceived value, while profit is still calculated using the true value.

This design creates uncertainty. If an agent overestimates an item, it may overbid and reduce profit. If it underestimates an item, it may miss a profitable opportunity. This allows the simulation to test how robust each strategy is when valuation information becomes unreliable.

C. Auction Memory

The memory component stores previous winning bids, losing bids, market prices and round history. It also calculates market signals such as average market price, volatility, pressure and competition level. These values are used by the Adaptive Agent to adjust future bids, which is related to learning from feedback [8].

Memory is mainly tested in Experiment 3, where the Adaptive Agent is compared with and without access to previous auction information.

D. Implemented Agents

All agents inherit from a shared BaseAgent class. This stores common data such as balance, wins, total spent, total profit, failed bids, bid history and profit history. It also provides shared methods for validating bids, recording results and calculating win rate.

Five concrete agent classes were implemented to match the strategies described in Section IV: Random, Conservative, Aggressive, Sniper and Adaptive. Each subclass defines its own bidding rule while reusing the shared validation and result-tracking methods from BaseAgent. The Adaptive Agent also stores aggressiveness history, allowing its learning behaviour to be analysed over time.

E. Simulation Runner and Result Logging

A simulation runner was created to automate repeated experiments. Each experiment was run multiple times so the results did not depend on one random simulation, which follows the use of repeated trials in agent market evaluation [6], [7].

The system records total profit, win rate, average bid, efficiency, failed bids and total wins. Results are exported as CSV files, with raw results stored separately from summary results. The analysis script then uses these CSV files to create comparison tables, bar charts, line graphs, error bars and the adaptive learning graph.

One implementation challenge was ensuring that the experiments remained fair despite randomness in item values, bidding order and agent decisions. To address this, repeated simulations were used and agent order was shuffled each

round to reduce ordering bias. Another challenge was separating perceived value from true value in the market noise experiment, so that agents made decisions using uncertain information while profit was still calculated using the actual item value. This was important because it allowed noise to affect agent behaviour without changing the objective evaluation metric.

VI. EXPERIMENTAL SETUP

The experiments evaluated agent behaviour under controlled auction conditions, similar to previous simulated bidding and trading agent studies [6], [7]. Each experiment was repeated multiple times to reduce the effect of randomness from item values, bidding order and agent decisions. Average results and standard deviations were recorded to measure both performance and stability.

Each simulation used 30 auction rounds. In each round, a new item was generated, agents submitted bids, the highest valid bidder won, and profit was calculated using the true item value minus the winning bid. Agents started with the same balance. The main metrics were total profit, win rate, average bid, bidding efficiency and failed bids.

Fixed random seeds were used in the final experiment scripts to make the results reproducible across reruns. This ensured that the same experiment settings could regenerate the same summary results and figures, while repeated runs were still used to calculate averages and standard deviations.

A. Experiment 1: Strategy Comparison

Experiment 1 compared the five main strategies: Random, Conservative, Aggressive, Sniper and Adaptive. The aim was to identify the general strengths and weaknesses of each strategy under the same market conditions.

B. Experiment 2: Number of Agents and Competition

Experiment 2 varied the number of agents across 5, 10 and 20 agents. This tested whether increasing competition reduced profit, efficiency and winning opportunities.

C. Experiment 3: Information Availability

Experiment 3 tested the Adaptive Agent with and without memory access. With memory, the agent could use previous winning bids, losing bids, market pressure, volatility and competition estimates. Without memory, it could not use previous outcomes. This tested whether extra market information improved adaptive decision-making [1], [8].

D. Experiment 4: Market Noise

Experiment 4 tested low, medium and high market noise. Noise changed the item's perceived value while keeping the true value unchanged. Agents made bidding decisions using perceived value, but profit was calculated using true value. This tested strategy robustness under uncertain valuation.

VII. RESULTS AND ANALYSIS

This section presents the results from the auction experiments. Performance is analysed using average profit, win rate, bidding efficiency and standard deviation. These metrics show not only which agents won more often, but also whether their wins were profitable and stable across repeated runs.

A. Experiment 1: Strategy Comparison

Experiment 1 compared the five bidding strategies under the same auction conditions.

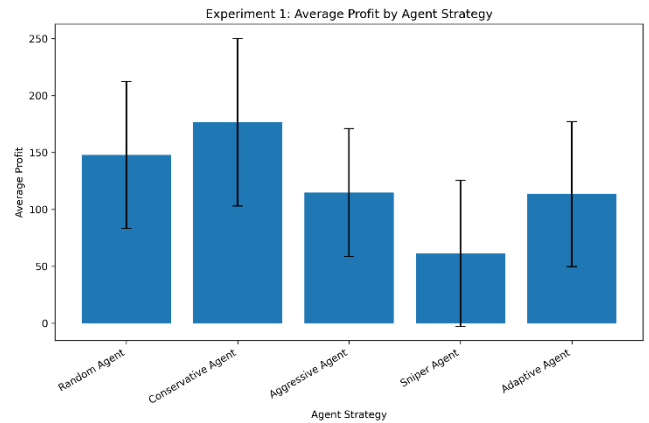


Fig. 1. Average profit achieved by each agent strategy across repeated simulation runs. Error bars represent standard deviation.

Figure 1 shows that the Conservative Agent achieved the highest average profit at **176.60**. This suggests that cautious bidding was effective because the agent protected its profit margins while still winning enough auctions to perform well. The Random Agent achieved the second highest average profit at **147.80**, followed by the Aggressive Agent at **114.90** and the Adaptive Agent at **113.50**. The Sniper Agent performed weakest, with an average profit of **61.40**, mainly because it entered auctions late and had fewer opportunities to place valid bids.

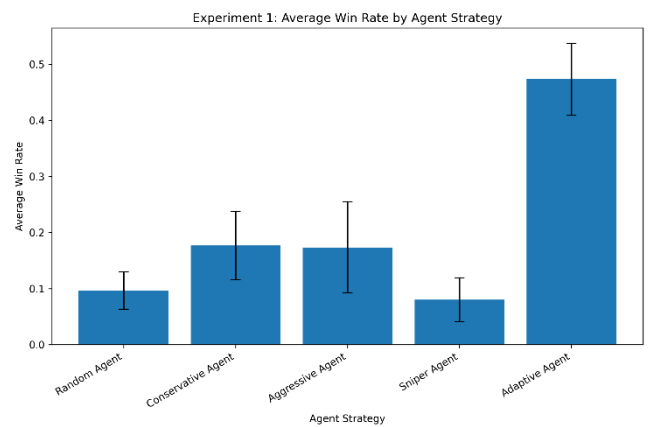


Fig. 2. Average win rate achieved by each agent strategy across repeated simulation runs. Error bars represent standard deviation.

Figure 2 shows that the Adaptive Agent achieved the highest win rate at **0.4733**. This means it won auctions more frequently than the other strategies. However, this did not lead to the highest profit. The Conservative Agent had a lower win rate of **0.1767** but achieved the highest average profit, showing that winning more auctions does not always mean performing better in a first-price auction [3], [4].

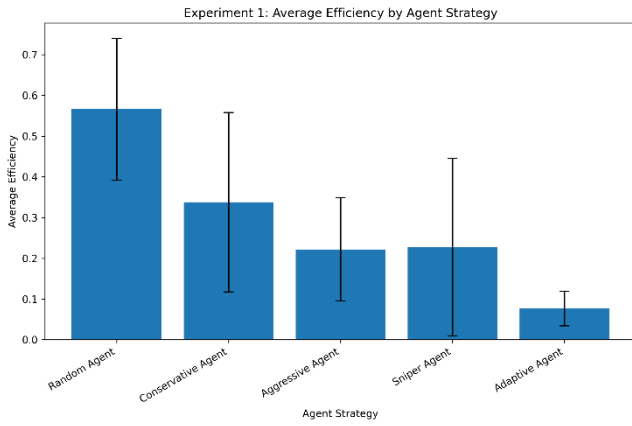


Fig. 3. Average bidding efficiency achieved by each agent strategy across repeated simulation runs. Error bars represent standard deviation.

Figure 3 shows that the Random Agent achieved the highest efficiency at **0.5662**. This does not mean it was the strongest overall strategy, because it had a low win rate. Instead, it suggests that when it did win, it sometimes won cheaply. The Conservative Agent also achieved strong efficiency at **0.3377**, while the Adaptive Agent had the lowest efficiency at **0.0766** despite having the highest win rate. This shows that the Adaptive Agent became highly competitive but often spent heavily to secure wins.

Overall, Experiment 1 shows that no single strategy dominated every metric. The Conservative Agent achieved the highest profit, the Adaptive Agent achieved the highest win rate, and the Random Agent achieved the highest efficiency. This supports the idea that auction agents should be evaluated using multiple metrics rather than win rate alone.

B. Experiment 2: Number of Agents and Competition

Experiment 2 tested how performance changed as competition increased. The number of agents was varied across **5, 10 and 20**, while the number of auction rounds stayed fixed at 30.

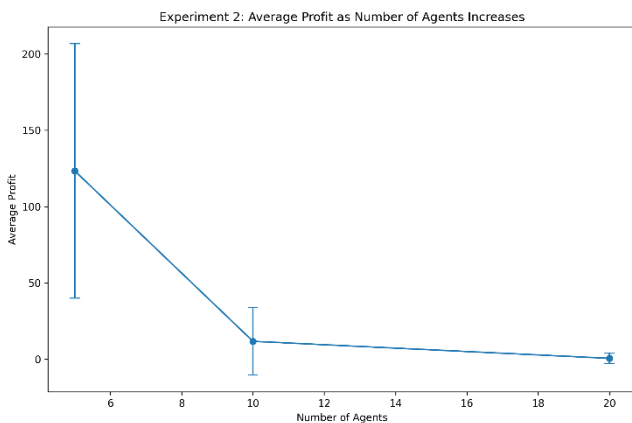


Fig. 4. Average profit as the number of agents increases from 5 to 20. Error bars represent standard deviation.

Figure 4 shows that average profit decreased sharply as the number of agents increased. With **5 agents**, average profit was **123.36**. This dropped to **11.71** with **10 agents**, and then to only **0.56** with **20 agents**. This shows that higher competition made profitable bidding much harder because more agents were competing for the same number of auction items [2].

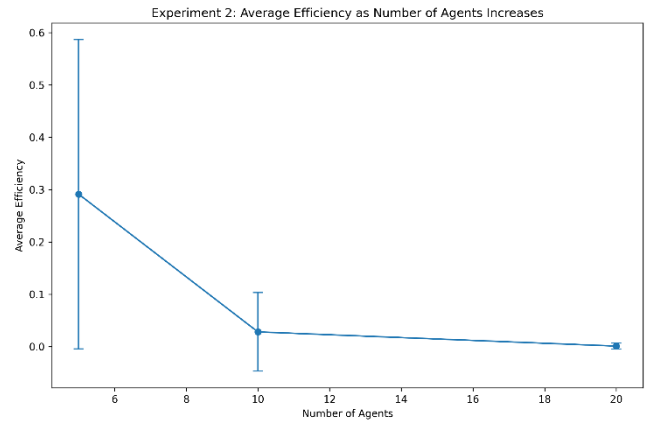


Fig. 5. Average bidding efficiency as the number of agents increases. Error bars represent standard deviation.

Figure 5 shows the same pattern for efficiency. Average efficiency decreased from **0.2912** with **5 agents** to **0.0282** with **10 agents**, before falling to **0.0011** with **20 agents**. This suggests that agents gained much less profit relative to spending as the market became more crowded.

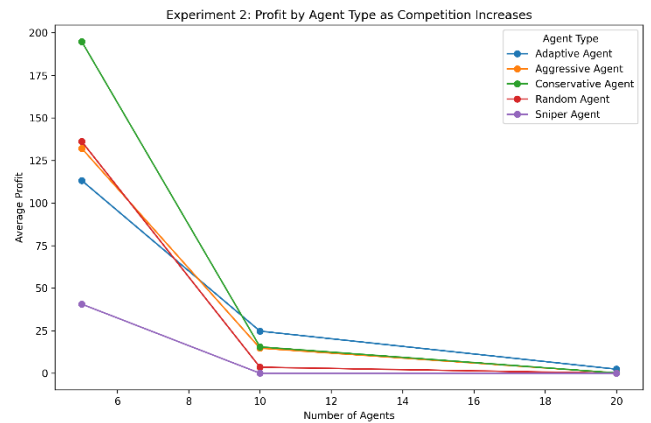


Fig. 6. Average profit by agent type as the number of agents increases.

Figure 6 shows that all strategies were negatively affected by increased competition. In the **5-agent** setting, agents were still able to generate meaningful profit, but performance dropped sharply when the market increased to **10 agents**. Under the **20-agent** condition, average profit was close to zero, showing that intense competition made it difficult for any strategy to perform well.

Overall, Experiment 2 shows that market structure strongly affects agent performance. A strategy that works in a smaller auction market may not remain effective when the number of competitors increases. As competition increased, profit, win rate and efficiency all declined.

C. Experiment 3: Information Availability

Experiment 3 tested whether auction memory improved the Adaptive Agent's performance. Two conditions were compared: **with memory** and **without memory**. Memory allowed the agent to use previous winning bids, losing bids, market average, volatility and competition pressure.

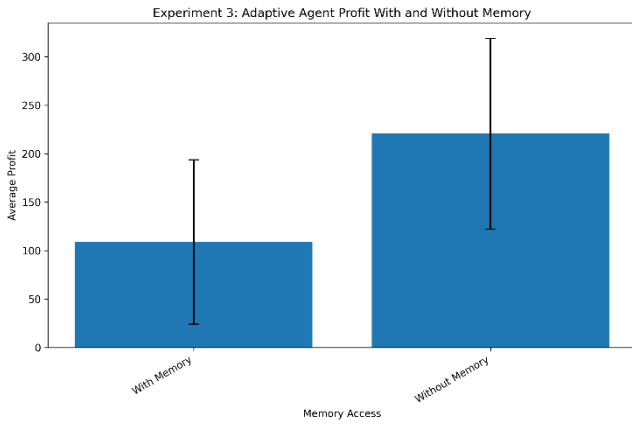


Fig. 7. Adaptive Agent average profit with and without memory access. Error bars represent standard deviation.

Figure 7 shows that memory access did **not** improve profitability. With memory, the Adaptive Agent achieved an average profit of **109.0**. Without memory, this increased to **220.6**. This suggests that the memory-based agent used the extra information to become more competitive, but not necessarily more profitable.

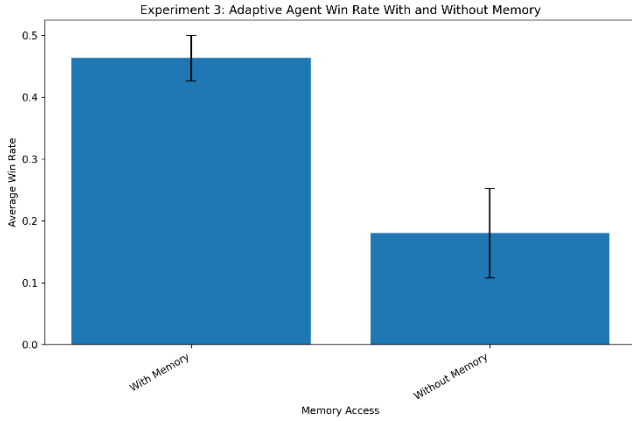


Fig. 8. Adaptive Agent average win rate with and without memory access. Error bars represent standard deviation.

Figure 8 shows that memory access improved win rate. With memory, the Adaptive Agent achieved a win rate of **0.4633**, compared with **0.1800** without memory. However, this higher win rate did not lead to higher profit. Since first-price auction winners pay their own bid, an agent can win frequently but still perform poorly if it overpays [3], [4].

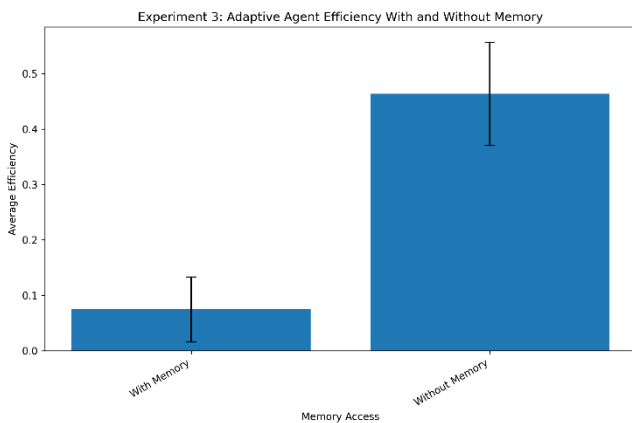


Fig. 9. Adaptive Agent average efficiency with and without memory access. Error bars represent standard deviation.

Figure 9 confirms this trade-off. With memory, efficiency was only **0.0746**, while without memory it was **0.4635**. This means the no-memory version gained more profit relative to spending. The memory-based agent became more aggressive after losses, which helped it win more often but also pushed bids closer to item value.

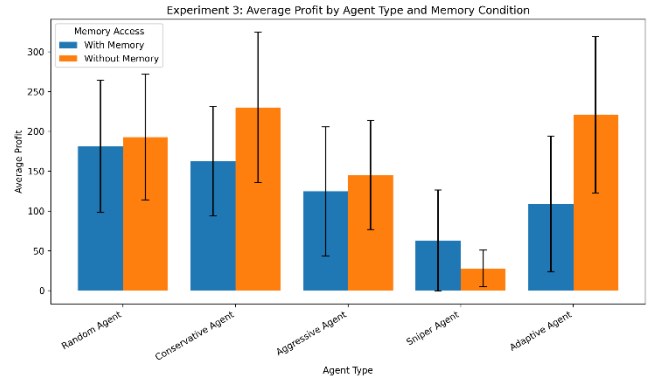


Fig. 10. Average profit by agent type under memory and no-memory conditions.

Figure 10 shows that the Adaptive Agent's memory condition also affected the wider auction environment. When memory was enabled, the Adaptive Agent became more competitive, which increased bidding pressure and influenced the performance of the other agents.

Overall, Experiment 3 shows that information availability produced a mixed result. Memory improved the Adaptive Agent's win rate, but reduced profit and efficiency. This suggests that additional information is only useful if the agent's decision rule uses it carefully. In this implementation, memory helped the agent respond to the market but also encouraged overbidding.

D. Experiment 4: Market Noise

Experiment 4 tested how market noise affected performance. Noise changed each item's perceived value while keeping its true value unchanged. Agents bid using perceived value, but profit was calculated using true value, testing decision-making under uncertainty [1], [2]. Three noise levels were used: **low**, **medium** and **high**.

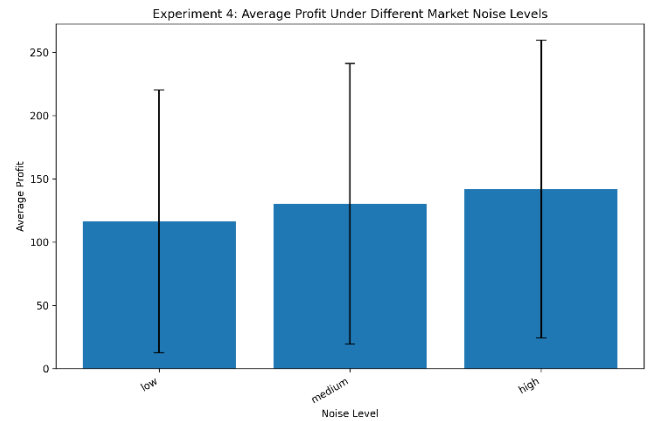


Fig. 11. Average profit under different market noise levels. Error bars represent standard deviation.

Figure 11 shows that average profit increased as noise increased. Average profit was **116.46** under low noise, **130.38** under medium noise and **141.94** under high noise. This suggests that noise did not simply reduce performance. Instead, it created more variation, sometimes allowing agents to win items at lower bids and gain higher profit margins. However, the high standard deviations show that noisy markets were less stable.

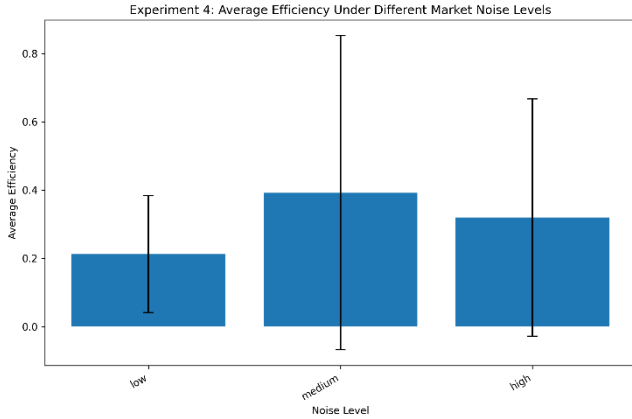


Fig. 12. Average efficiency under different market noise levels. Error bars represent standard deviation.

Figure 12 shows that efficiency was highest under medium noise. Efficiency increased from **0.2123** under low noise to **0.3919** under medium noise, then decreased to **0.3199** under high noise. This suggests that moderate uncertainty created some profitable opportunities, while excessive noise made valuation less reliable.

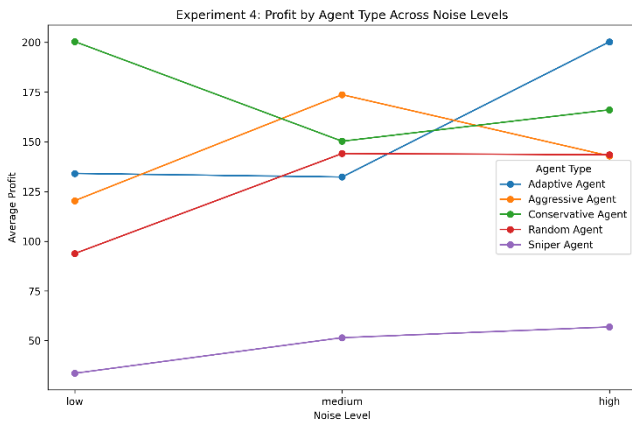


Fig. 13. Average profit by agent type across different market noise levels.

Figure 13 shows that different strategies responded differently to noise. The Adaptive Agent performed best under high noise, increasing from **134.1** under low noise to **200.3** under high noise. The Conservative Agent performed best under low noise, with **200.4**, suggesting that cautious bidding worked better when values were stable. The Aggressive Agent performed best under medium noise, while the Sniper Agent remained the weakest strategy across all noise levels.

Overall, Experiment 4 shows that uncertainty created both opportunities and risks. Noise sometimes improved profit, but it also made outcomes less predictable. This supports the idea that autonomous bidding agents must be evaluated not only by average performance, but also by stability under uncertain market conditions.

VIII. DISCUSSION

The experiments show that agent performance was affected by both strategy design and market conditions, which reflects the importance of environment design in multi-agent systems [1], [2]. Although all agents followed the same auction rules, they behaved differently because each strategy made different assumptions about risk, timing, competition and information. The main finding is that success in an auction system is not only about winning items, but about winning them efficiently.

A. Overall Strategy Performance

Experiment 1 showed that no single strategy dominated every metric. The Conservative Agent achieved the highest average profit because cautious bidding protected profit margins. The Adaptive Agent achieved the highest win rate, showing that its memory-based behaviour made it highly competitive, but this did not produce the highest profit or efficiency. The Random Agent achieved the highest efficiency because it sometimes won cheaply, while the Sniper Agent performed weakest due to fewer valid bidding opportunities. Overall, successful bidding required a balance between competitiveness and controlled spending.

B. Effect of Competition

Experiment 2 showed that increasing the number of agents sharply reduced profit and efficiency. With more agents competing for the same number of items, each agent had fewer opportunities to win and stronger pressure to bid higher. This supports the idea that multi-agent outcomes depend heavily on the behaviour of other competing agents [2]. Under 20 agents, profit fell close to zero, showing that a strategy that works in a small market may fail in a crowded one. This suggests that market structure can be as important as agent design.

C. Effect of Information Availability

Experiment 3 showed that memory access improved the Adaptive Agent's win rate but reduced its profit and efficiency. Memory made the agent more competitive after losses, causing it to bid closer to item value. This shows that more information does not automatically improve performance. Adaptive behaviour depends on how feedback is interpreted and applied [8]. In this project, the learning rule was too focused on winning rather than profit.

D. Effect of Market Noise

Experiment 4 showed that market noise created both opportunities and risks. Since agents bid using perceived value but profit was calculated using true value, inaccurate valuation could either help agents win cheaply or cause them to overpay. The Adaptive Agent performed strongly under high noise, but its efficiency remained relatively low. This shows that noisy markets can increase opportunity but also make outcomes less predictable.

E. Adaptive Behaviour and Learning

The Adaptive Agent demonstrated a simple form of learning by adjusting its aggressiveness using previous outcomes [8].

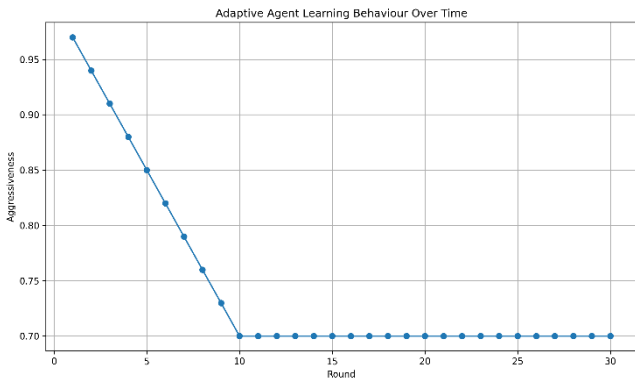


Fig. 14. Adaptive Agent aggressiveness over 30 auction rounds.

Figure 14 shows that its aggressiveness decreased from 0.97 to 0.70 by round 10, then stayed stable. This shows that the agent changed its behaviour in response to stored auction outcomes, but also that the learning rule was limited. Overall, the Adaptive Agent revealed the trade-off between learning, competition and profitability, suggesting that future adaptive agents should learn from profit rather than only wins and losses.

IX. CONCLUSION

A. Summary of Findings

This project investigated autonomous bidding agents in a simulated multi-agent auction market. Five strategies were implemented: Random, Conservative, Aggressive, Sniper and Adaptive. The experiments tested strategy performance, competition, information availability and market noise.

The results show that agent performance depends on both the bidding strategy and the market environment. In the strategy comparison experiment, the Conservative Agent achieved the highest average profit, while the Adaptive Agent achieved the highest win rate but lower efficiency. This shows that winning more auctions does not always mean better performance.

The information availability experiment further supported this finding. Memory increased the Adaptive Agent's win rate but reduced its profit and efficiency because it encouraged more competitive bidding. This shows that additional information does not automatically improve performance unless the agent's decision rule uses it effectively.

The experiments also showed that competition strongly reduced profit and efficiency. As the number of agents increased, average profit dropped sharply. Market noise created both opportunities and risks because agents made decisions using perceived values while profit was calculated using true values. Overall, the project shows that autonomous bidding agents can be evaluated effectively through simulation [6], [7], and that agent success should be judged using profit, win rate, efficiency, stability and robustness together.

B. Reflection

This project showed that both agent design and environment design are important in intelligent agent systems. A key lesson was that one metric is not enough to evaluate performance. For example, the Adaptive Agent often won more auctions when memory was enabled, but this did not always make it more profitable. Using multiple metrics such

as profit, win rate, efficiency and failed bids made the analysis more balanced.

The Adaptive Agent was also useful because it produced imperfect but meaningful behaviour. Its memory-based learning was simple, but it created realistic trade-offs between competitiveness and overbidding. This gave the project stronger results to discuss than if the adaptive strategy had simply performed best in every condition.

C. Future Work

Future work could improve the Adaptive Agent by making it learn from profit rather than only from wins and losses. This would help it balance competitiveness with efficiency. A more advanced version could also use reinforcement learning, where the agent learns a bidding policy through repeated interaction and reward feedback [8]. The environment could also be extended with different auction formats, changing item supply, reserve prices, varied budgets and different risk preferences. Finally, running more simulations and analysing profit per round, bid trends and balance changes would make the evaluation more reliable.

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